

Reliable Offline Tracking for Long-Shot Windsurfing Video

Domain-specific pose, appearance, and global association

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Abstract

This report describes the engineering decisions behind a system that turns long-shot, shore-recorded windsurfing footage into identity-consistent rider tracks and stable focused views. The central challenge is not object count but error tolerance: a single identity switch can contaminate the extracted video of a rider. Large camera pans, small targets, prolonged co-visibility, and occlusions make generic online tracking a poor fit. We therefore combine a domain-specific two-keypoint pose model, sail-color appearance descriptors, camera-compensated motion, and offline global tracklet linking.

1 Problem Definition and Operating Regime

1.1 The useful output is an identity-consistent video

The input is long-shot windsurfing footage recorded from shore. A typical scene contains only a handful of surfers, so detection density is modest compared with crowded pedestrian tracking. The difficulty lies elsewhere: the subjects are small, several sails can remain close or overlap for extended periods, and a rider may be partly or fully occluded before reappearing. Meanwhile, the operator follows the action with large pans—about 90° in demanding examples, and sometimes more. Almost every background feature then moves in image coordinates, while the visible background itself changes over the course of the shot.

The desired result is not merely a plausible bounding box in most frames. It is a separate, continuous trajectory for each physical surfer, suitable for extracting a focused rider video. If a track changes identity once, the resulting video jumps to another rider and can no longer serve its purpose. Identity consistency is therefore the primary objective; smooth localization and high detection recall matter only after that constraint is met.

We consequently treat association errors asymmetrically. A missed link produces two fragments that can be inspected or joined later. A false link contaminates an otherwise valid trajectory and may invalidate the complete extracted sequence. In ambiguous cases, fragmentation or abstention is preferable to an unjustified merge. This is a design requirement—effectively zero tolerance for identity switches—not a claim that the present system achieves perfect tracking on arbitrary footage.

1.2 Why ordinary online tracking is a poor fit

General-purpose multi-object trackers commonly process frames causally: they predict each active track into the next frame, assign the current detections, and carry that decision forward. This is attractive for real-time operation, but it commits at the moment evidence is weakest. During a crossing or occlusion, two continuations can be equally plausible from past motion alone; evidence that appears several seconds later cannot undo an early identity switch reliably.

Raw image-space motion is also a misleading cue in this regime. During a pan, a stationary background point and every surfer receive a large common displacement. The residual motion of an individual surfer may be much smaller than this camera component. As-

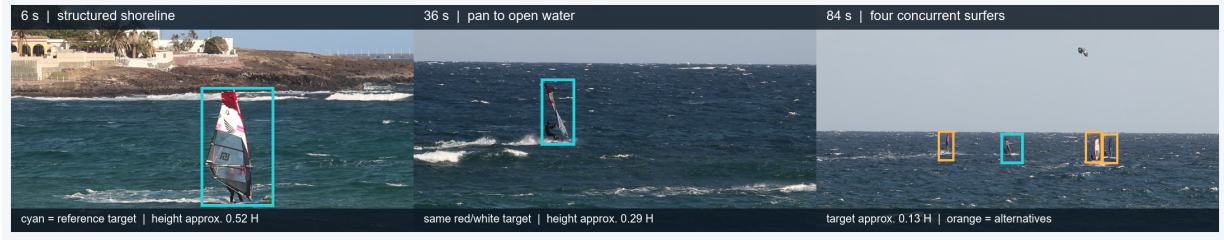


Figure 1: Representative frames from MVI_5347.MP4 at 6, 36, and 84s. The cyan annotation follows the same red-and-white sail as the camera moves from shoreline to open water; its image height changes from approximately $0.52H$ to $0.13H$. At 84s, three other surfers (orange) are visible alongside it as plausible association candidates. The composite illustrates changing background, scale variation, and identity ambiguity within one shot.

sociation therefore depends on estimating global camera motion and applying it to the motion model before comparing a prediction with a detection. The production pipeline makes this coupling explicit: it estimates masked frame-to-frame transforms while excluding detected surfers, then passes those transforms to both the greedy tracklet builder and the global linker (`video_processing/main_tracking.py`).

There is no real-time requirement for the final analysis: the complete recording is available before a focused video is generated. The system uses that freedom in two stages. A conservative causal pass accepts only clear local matches and leaves ambiguous observations as separate tracklets. An offline integer programming pass then evaluates feasible links between those tracklets jointly, using evidence on both sides of a gap. Its graph includes the option to begin or end a trajectory instead of forcing a link, and can discard very short suspected false-positive fragments in the production tracking implementation. Thus future evidence informs association without weakening the conservative local decisions that protect track purity.

The problem is therefore narrower than general multi-object tracking, but its loss function is harsher: few objects, difficult camera motion, long periods of ambiguity, and an output that is useful only while identity remains intact. The remainder of this report explains how pose, sail-specific appearance, camera compensation, and offline global association were chosen around that operating regime.

2 Pose as a Stable Virtual Camera Signal

Tracking answers *which* surfer is present in each frame, but a useful focused view also needs to decide where to place the virtual camera and how tightly to crop. A detector bounding box is a poor control signal for this purpose. A windsurf rig is articulated, strongly anisotropic, and continually rotates with respect to the camera. Its axis-aligned box can therefore change width, height, and centre even when the rider has moved little and remains at approximately the same distance. Using the box centre and height directly makes the crop drift across the subject and causes visible zoom pumping.

2.1 Two semantic points instead of a generic human pose

The production detector is a YOLO pose model supervised for one class and two domain-specific keypoints: the boom–mast junction and the mast tip. These points were chosen for the rendering geometry rather than to describe the complete human body. The boom–mast junction gives a repeatable reference near the rider and sail, while the distance from that junction to the mast tip provides a physical scale cue that is substantially less sensitive than an axis-aligned box height to in-plane rotation.

The labels build on the manually reviewed detection boxes. A dedicated annotation tool presents an enlarged crop for each box, permits either keypoint to be marked as not visible, and writes two-keypoint YOLO pose labels. The checked-in pose project contains 1,123 manually labelled training images and 56 validation images. The workflow also supports conservative pseudo-labelling: a prediction is accepted only after matching it to an existing box and applying detection, overlap, keypoint confidence, and spatial plausibility gates; pseudo-labels remain separate from manual labels and are included in training only when requested. Finally, a hard-window miner runs the complete local pipeline and ranks short intervals by keypoint jitter relative to the rendered anchor. This directs subsequent review towards failures that are visually disruptive in the final crop, rather than towards another uniform sample of easy frames.

2.2 From keypoints to crop centre

For a detection with bounding box top y_1 , height h , and centre (c_x, c_y) , the fallback anchor is

$$\mathbf{a}_{\text{box}} = (c_x, y_1 + 0.6h). \quad (1)$$

Placing it below the geometric box centre is a deliberate bias towards the rider and board rather than the empty upper part of a tall sail box. When the boom–mast junction $\mathbf{b} = (b_x, b_y)$ is visible, the implementation first constructs a low-noise proxy for the mast tip, $\tilde{\mathbf{t}} = (b_x, y_1)$, and places the semantic anchor at

$$\mathbf{a}_{\text{pose}} = 0.15\tilde{\mathbf{t}} + 0.85\mathbf{b}. \quad (2)$$

Thus the crop is anchored slightly above the boom junction. The measured mast-tip coordinate is intentionally not used for position: its larger lever arm would turn localization noise into unnecessary crop motion. It remains valuable for scale.

Small detections do not contain enough pixels for equally reliable keypoint localization. The final unsmoothed anchor therefore blends $\mathbf{a}_{\text{pose}}$ with $\mathbf{a}_{\text{box}}$ according to the shorter box dimension. The pose weight rises linearly from zero at 45 pixels to one at 55 pixels. A keypoint is considered visible at confidence 0.15 or greater; below that threshold, the box fallback is used.

Missing observations are handled as transitions, not abrupt mode switches. Gaps of up to five frames between visible boom keypoints are interpolated directly between semantic anchors. Across longer gaps, the anchor eases from the last semantic point to the per-frame box fallback and eases back when a reliable keypoint returns. Leading and trailing missing spans receive analogous fallback or short hold behaviour. A final five-frame robust weighted mean suppresses isolated jumps while giving the current frame the largest weight; Huber reweighting limits the influence of spatial outliers. This processing operates on the dense tracks produced by the tracking post-processor, so every frame in a track receives an anchor even when its detection was interpolated.

2.3 A scale tied to the rig

When both keypoints are visible, the scale signal is the Euclidean length $\ell = \|\mathbf{b} - \mathbf{t}\|_2$ of the observed boom-to-tip mast segment. Short missing spans are interpolated and the last valid value is held for up to five frames; otherwise the vertical distance from the current anchor to the box top supplies a fallback. A centred moving mean with radius 15 frames smooths this length signal. The requested crop height in pixels is then

$$C = \max\left(\frac{\bar{\ell}}{0.75}, \frac{h}{0.9}, 1\right), \quad s = \text{clip}\left(\frac{C}{H}, 10^{-6}, 1\right), \quad (3)$$

where H is the source-video height. The first term aims to make the semantic mast segment occupy 75% of the output height; the second prevents the detected box from being cropped more tightly than a 90% fill. Taking the maximum preserves context when either signal calls for a wider view.

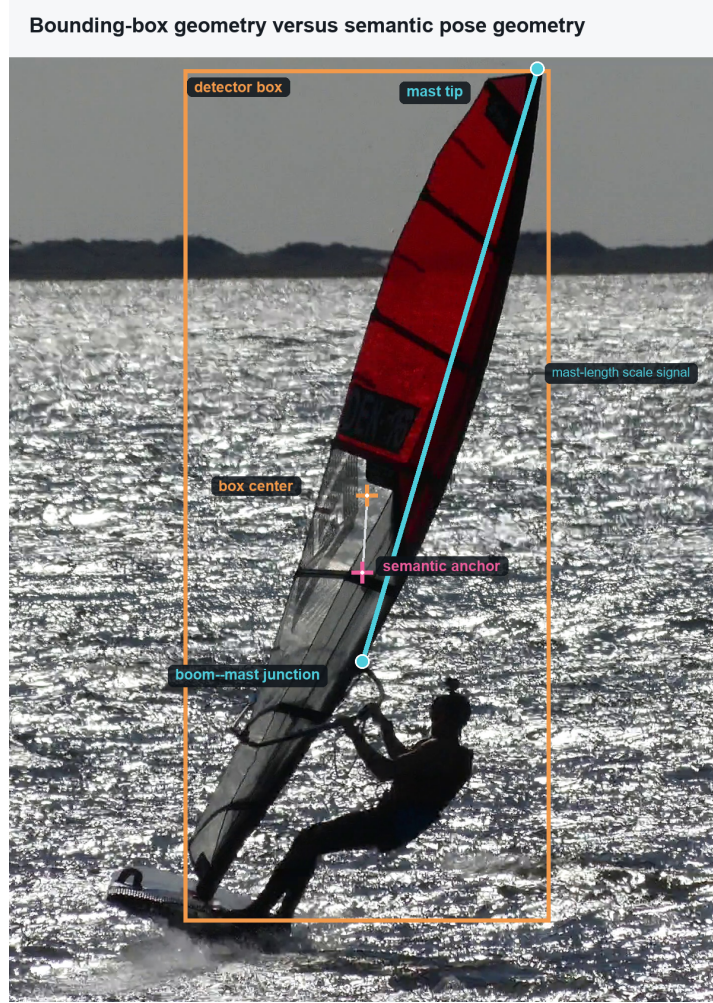


Figure 2: Pose-derived virtual-camera geometry on a representative labelled frame. The detector box and its centre (orange) describe image extent. The boom–mast junction and mast tip (cyan) instead provide a rider-relative anchor (magenta) and a mast-length scale signal. Production rendering smooths these signals over time and falls back to box geometry when keypoints are unavailable.

The resulting normalized anchor and scale are stored with every renderable detection and consumed directly by the focused player. They stabilize the *composition* of an individual rider: which semantic point stays near the centre and how large the rig appears. This is distinct from global camera-motion compensation. The latter estimates frame-to-frame background transforms (while masking detected surfers), supports motion-based association, and supplies an optional stabilized overview. It does not replace the pose-derived virtual camera; a globally stabilized frame can still contain a rotating rig whose bounding box centre and dimensions fluctuate.

3 Domain-Specific Offline Multi-Object Tracking

The tracker is organized around the asymmetric error cost introduced in section 1. It is acceptable to leave a difficult interval as two fragments; it is not acceptable to join fragments belonging to different surfers merely to increase continuity. The production pipeline therefore separates association into two levels:

$$\text{detections} \longrightarrow \text{high-purity local tracklets} \longrightarrow \text{globally linked trajectories.} \quad (4)$$

The first stage resolves only short, unambiguous transitions. The second can inspect all tracklets in the recording before deciding which gaps to bridge.

3.1 Why general-purpose components were not sufficient

The project investigated OC-SORT [1] and BoT-SORT [2], including variants with camera-motion compensation and appearance features. It also implemented appearance backends based on OSNet person re-identification and a general Vision Transformer. These are useful reference designs, but their usual operating assumptions do not match this narrow application particularly well. A causal tracker must commit while two sails overlap or cross; subsequent frames may reveal the correct continuation, but arrive too late to prevent the original identity switch. Moreover, a person-ReID network devotes its capacity to cues such as clothing and body appearance. In long-shot windsurfing footage, the person occupies very few pixels, whereas the sail is large, deliberately colorful, and often visually distinctive.

This experience does not imply that the custom tracker is generally superior to OC-SORT, BoT-SORT, or learned ReID. The comparison is specific to shore-recorded windsurfing, offline processing, and a loss function dominated by even a single false merge. The implemented comparison and optimization tools document that the alternatives were tested, but the current benchmark harness has drifted from the production embedding and tracker interfaces. We therefore avoid a numerical outperformance claim until the baselines have been rerun on aligned inputs and representative manually verified sequences.

3.2 Sail color as an appearance descriptor

The production appearance backend replaces generic semantic features with a compact descriptor tailored to the most stable visible property in this domain: sail color. For every detection crop it computes a joint histogram of the chromatic Lab channels (a, b), a circular hue histogram, and a lower-weight luminance histogram. Pixel contributions are weighted by S^γ , where S is HSV saturation, so strongly colored sail panels contribute more than grey water, spray, or haze. Hue votes are linearly shared between adjacent circular bins, avoiding a discontinuity at the hue wraparound.

Color alone would discard useful coarse layout. The crop is consequently divided into three bands along its vertical axis, with a fourth block covering the complete crop. The Lab, hue, and luminance histograms are concatenated for each block. Every histogram first receives an L1 normalization and square-root mapping; the concatenated vector is then L2-normalized. The resulting representation supports a bounded Hellinger-style distance through the cosine similarity of the transformed vectors. Tracklet appearance prototypes are exponential moving averages, which reduce the effect of individual blurred, partially occluded, or poorly exposed crops.

The descriptor also records a useful negative result. An earlier implementation estimated foreground from the color statistics of the crop border, using a robust Lab background model, a smoothness gate, and connected components. In the active path this is bypassed by a simple mask that removes the outer 10% of the crop. The fixed centre margin is less ambitious, but removes the descriptor’s dependence on a background model as the scene changes from beach to water or a sail touches the bounding-box edge. This is an example of domain adaptation through simplification: the descriptor need not segment the complete rig if its central, saturated color distribution is already sufficiently discriminative.

3.3 Conservative formation of local tracklets

Each detection initially forms a one-observation track. A greedy preprocessor then processes frames in temporal order and maintains, for every active tracklet, a constant-velocity Kalman state and an exponentially averaged appearance descriptor. Before scoring a detection, the



Figure 3: Illustrative production-descriptor comparison on three consecutive samples from one project sequence. Repeated crops of each sail remain similar, while the black-and-white and red sails remain distinguishable. Values are mean descriptor similarities over the displayed crops, not calibrated probabilities or benchmark accuracy. The example is intentionally small and hand-selected; it explains the cue used by the tracker rather than evaluating it.

Kalman state is advanced through the measured frame-to-frame camera transforms. Thus the expected box centre follows both the surfer’s estimated velocity and the image displacement induced by camera pan and rotation.

Candidate transitions are gated by two probabilities: one derived from the position-space Mahalanobis distance of the camera-compensated prediction, and one derived from sail-appearance distance. Strict matches require both cues to pass high thresholds and may span at most six frames. Looser candidates are retained only when sufficiently supported by one strong cue or by both weaker cues. Crucially, the preprocessor does not break ties by choosing the numerically best ambiguous option. A detection is appended only when it has one admissible candidate under the clean-first rule. Multiple candidates, multiple detections proposing the same tracklet, or overlap with a newly created track cause a split and fresh tracklets. This deliberate strategy trades recall for purity: uncertainty is preserved for the offline stage instead of being hidden inside an irreversible local decision.

3.4 Camera-compensated candidate costs

Let T_i and T_j be non-overlapping tracklets with T_i ending before T_j begins. The offline tracker creates a directed candidate edge $i \rightarrow j$ only when the temporal gap is non-negative and no longer than the configured horizon, currently three seconds. A Kalman filter is fitted once to all detections of T_i . Its end state is then advanced to the first observations of T_j , applying the estimated camera transform at every intervening frame. This matters during long pans: without compensation, the shared movement of the entire image would dominate the residual motion that distinguishes one surfer from another.

Every candidate receives a weighted negative-log-likelihood cost

$$C_{ij} = w_m C_{ij}^{\text{motion}} + w_a C_{ij}^{\text{appearance}} + w_g C_{ij}^{\text{gap}}. \quad (5)$$

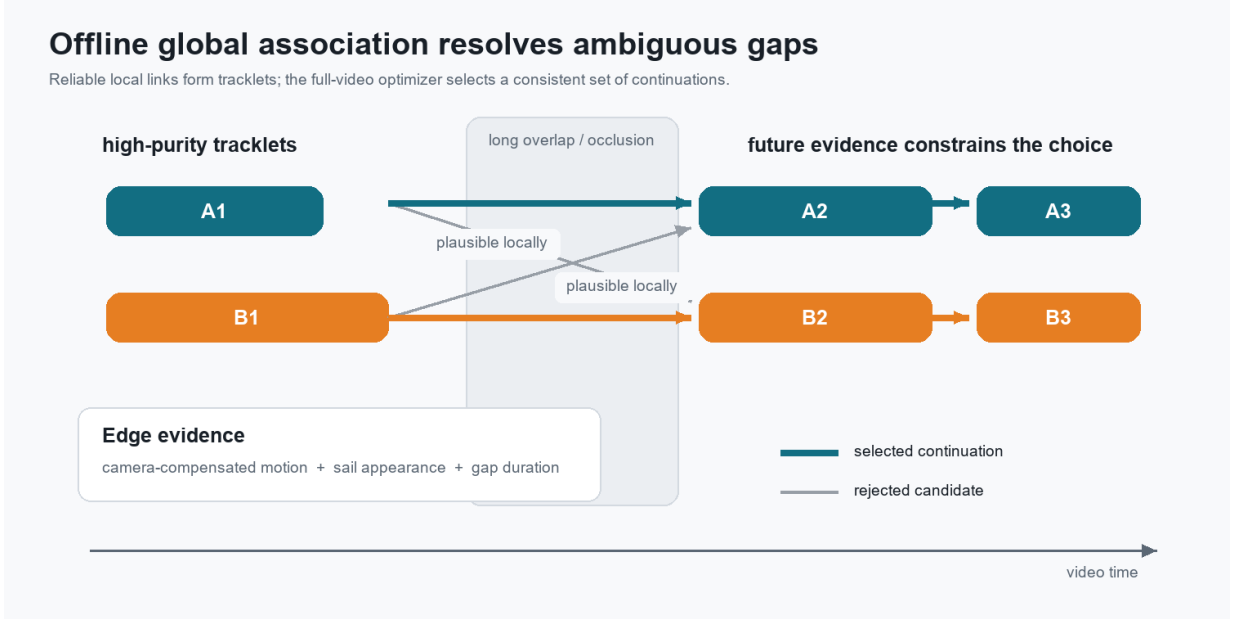


Figure 4: Conceptual view of offline tracklet linking. The conservative local pass preserves ambiguity across a long overlap or occlusion. With complete-video evidence available, the optimizer selects a mutually consistent set of continuations using camera-compensated motion, sail appearance, and gap duration. The diagram omits start, end, and discard variables for clarity.

The motion term averages $-\log p$ values obtained from the Mahalanobis distance to the first few detections of T_j . The appearance term compares an exponentially weighted prototype at the end of T_i with a reverse-weighted prototype at the beginning of T_j and maps the Hellinger-style distance to a match probability. The gap term is

$$C_{ij}^{\text{gap}} = -\Delta f_{ij} \log p_{\text{miss}}, \quad (6)$$

so a proposed link must accumulate stronger motion and appearance evidence as the number Δf_{ij} of unobserved frames grows. The fitted weights calibrate the relative influence of these three terms for the project data; they should not be interpreted as universal MOT constants.

3.5 One global decision over the recording

The candidate graph is solved as a minimum-cost integer program. For each tracklet i , binary variables represent starting a trajectory at i , ending at i , linking i to a candidate successor, or—for eligible short fragments—discarding i . The constraints require exactly one incoming choice and exactly one outgoing choice for every retained tracklet. The selected edges therefore form disjoint, time-ordered paths rather than competing local assignments.

Starting and ending are explicit alternatives to linking. Their costs are reduced when a tracklet begins or ends within the outer 10% of the image, because a surfer entering or leaving the field of view is more plausible there than in the centre. An edge is removed before optimization whenever linking i to j is no cheaper than ending i and starting j independently. The graph is further limited to the best few outgoing candidates per node. These operations reduce solver work while preserving the intended decision boundary: a weak link should become a fragmentation, not an identity switch.

Very short detector artifacts pose a separate problem. For fragments up to a configured number of detections, the optimizer exposes a discard option whose cost grows exponentially with fragment length. A one-frame false positive can therefore be removed without forcing it to become a trajectory or attach to a real surfer, while discarding a sustained observation rapidly becomes unattractive and is eventually disallowed.

The advantage of this formulation is non-causal rather than mystical “future knowledge.” A difficult edge is judged using a stable tracklet before the gap, a stable tracklet after it, and the other feasible links throughout the recording. The integer program selects these alternatives jointly, so two locally plausible continuations cannot both claim the same successor. The current production pipeline uses this single-pass `ILPTracker`; the older iterative ILP implementation remains an experiment and is not the authoritative execution path.

4 Experiments, Evaluation, and Failure Analysis

The evaluation problem follows directly from the product requirement. Conventional multi-object-tracking scores are useful summaries, but an average that tolerates an occasional identity switch is not sufficient here: one false merge can make the complete focused video of a rider unusable. We therefore used quantitative association tests to guide development, while retaining frame-accurate visual review as the final check on complete trajectories. This section distinguishes those established evaluation mechanisms and retained diagnostic results from comparative claims that would require a fresh benchmark run.

4.1 Golden tracklet associations

The project includes an annotation tool for constructing “golden” associations from the same detections presented to the tracker. It first runs the conservative preprocessor described in section 3; an annotator then assigns every resulting tracklet either to a physical surfer identity or to the discard class. Tracklets assigned to one identity are forbidden from occupying the same frame. The saved record retains the source detection coordinates and their human-assigned identity, so competing association methods can be scored over an identical universe of observations.

This protocol deliberately evaluates association separately from detection. It asks whether the system grouped the available observations correctly, not whether YOLO found every surfer in the source video. That separation made tracker iteration much faster and avoided attributing a missed detection to the association algorithm. It also means that an association score alone is not an end-to-end tracking score; detection omissions, interpolation quality, and the usefulness of the final crop must be reviewed separately.

For N retained detections, the evaluator considers all unordered pairs. A pair is positive when both detections belong to the same annotated surfer. In this view,

$$P_{\text{pair}} = \frac{\text{correct same-identity pairs}}{\text{all pairs joined by the tracker}}, \quad (7)$$

$$R_{\text{pair}} = \frac{\text{correct same-identity pairs}}{\text{all annotated same-identity pairs}}. \quad (8)$$

Pairwise precision exposes false merges, whereas pairwise recall exposes fragmentation. Their F1 score, together with the Rand and same-pair Jaccard indices, was used as a compact clustering measure. Missing predicted observations can be treated as unique singleton tracks, preventing missing data from being rewarded as one artificial cluster. Discard decisions are evaluated independently as a binary classification problem.

The distinction between precision and recall is essential for this application. A high pairwise F1 value can still conceal the one identity switch that invalidates a long output. Model selection should therefore treat pairwise precision as a hard or near-hard constraint, inspect the absolute number of false joins, and only then use recall to choose among sufficiently pure configurations. The final acceptance test remains a complete-video audit in which every trajectory is followed through crossings, long overlaps, occlusions, and camera pans.

4.2 Optimization as structured engineering evidence

The repository records a staged optimization workflow rather than a single opaque parameter search. Detector confidence and non-maximum-suppression thresholds can be tuned against labelled boxes under an explicit false-positive-per-image constraint. Kalman process and measurement noise are fitted by minimizing one-step bounding-box prediction error on golden trajectories after camera compensation. Motion and appearance mappings are then calibrated on positive fragment pairs sampled from one identity and negative pairs sampled from different identities. Finally, Optuna-based search evaluates preprocessor and ILP parameters using pairwise association scores; trial-invariant embeddings and camera transforms are cached, and videos can be evaluated in parallel.

This sequence was valuable even where it did not produce a publication-style benchmark. It separated failures due to localization, motion prediction, appearance separation, local stitching, and global linking. Interactive threshold tools and the golden-tracklet viewer complemented the scalar objectives by exposing the cases that matter disproportionately to the product, especially a plausible but wrong merge. The fitted settings are consequently best understood as engineering choices for this camera geometry, subject scale, and collection of sails—not as universal tracker hyperparameters.

4.3 A retained synthetic motion diagnostic

Camera compensation can be checked independently because a known perturbation can be injected into a video. The project contains a generator that adds controlled frame-wise translation and rotation and an evaluator that subtracts the motion estimated on the unperturbed source before comparing the recovered perturbation with ground truth. One retained 60-frame translation-only run is summarized in table 1.

Table 1: Recovery error on one 60-frame synthetic translation test (59 frame-to-frame transforms). Independent random offsets of up to ± 10 px per axis were added with seed 1; source-video motion was estimated separately and subtracted. Lower is better.

Method	RMSE			MAE		
	x [px]	y [px]	θ [mrad]	x [px]	y [px]	θ [mrad]
VidStab	0.257	0.184	0.153	0.189	0.140	0.118
GMC	0.083	0.086	0.044	0.065	0.069	0.033

The test injected translation only ($\theta = 0$); angular entries therefore measure spurious estimated rotation. This diagnostic validates the implementation on one short synthetic clip and is not evidence of real-video tracking accuracy or general superiority.

On this diagnostic, the global-motion-compensation (GMC) implementation had lower translation error than the tested VidStab configuration. The result is useful as a unit-scale validation that the estimator recovers known motion and does not invent substantial rotation. It is intentionally not promoted to a real-world performance claim: the run contains only 59 transitions from one synthetic clip, injects no rotation, and does not reproduce a long pan with a changing background. A reportable camera-motion benchmark should add multiple source scenes, rotations and scale changes, foreground-mask ablations, feature-poor water backgrounds, and failure rates for transforms that cannot be estimated.

4.4 Qualitative failure analysis

The most consequential failures occur when several weak cues coincide. During a long overlap, motion cannot separate two riders; if the sails also share similar colors or one crop is dominated by spray and background, appearance may not resolve the continuation either. Rapid pans can

reduce the number of background features shared by adjacent frames, while small or blurred detections make both the Kalman update and color descriptor less reliable. Leaving the observations fragmented is the intended safe response in these cases. A false association is the more serious failure and must be identified by inspecting the selected ILP edge and the crops on both sides of the gap.

Other failures affect presentation rather than identity. Missed detections create interpolated boxes; extended gaps eventually make those boxes physically untrustworthy. Occluded or very small keypoints cause the focused view to fall back to bounding-box geometry, potentially reintroducing crop drift or scale pumping. Hard-window mining in the pose workflow ranks intervals with unstable anchors so that these visible defects can be reviewed and added to the labelled set. For global motion, foreground masking reduces the risk that a large moving sail is mistaken for the background, but it cannot create reliable correspondences when the remaining scene is textureless or newly revealed by a pan.

4.5 Scope of the present evidence

The codebase contains implementations and comparison hooks for OC-SORT, BoT-SORT, greedy association, and several ILP variants, documenting that general-purpose alternatives were explored. However, no current, reproducible output table compares those baselines with the production pipeline on a held-out set. The comparison harness also predates the active color-stripe embedding and single-pass `ILPTracker` interfaces, and some historical golden files require compatibility loading after refactors. Reporting the old runs as a current leaderboard would mix different inputs and implementations.

The supported conclusion is therefore narrower: those experiments revealed failure modes that motivated a domain-specific, offline design, and the present tooling makes a controlled comparison possible after the baselines are aligned. A release-quality evaluation should freeze representative train/tuning and held-out videos; report pairwise precision and recall, false-merge counts, fragmentation, discard quality, runtime, and complete-track pass/fail rates; and stratify sequences by pan magnitude, overlap duration, target size, and appearance ambiguity. The intended claim is improved reliability on this specific long-shot windsurfing task under a near-zero identity-switch objective—not general superiority over state-of-the-art MOT.

5 Delivering Frame-Accurate Results in the Browser

The tracking result is indexed by frame: detections, stabilization transforms, pose anchors, and user annotations must all refer to the image that is actually displayed. An ordinary HTML video element exposes media time rather than a stable frame identity. Seeking to a time may restart decoding at an earlier keyframe, and presentation timestamps need not provide a unique ordering by themselves. This is sufficient for ordinary playback, but it is a poor contract for single-frame stepping, hit-testing a small detection, or checking the exact transition across an occlusion.

For supported browser and codec combinations, the project therefore uses an explicit WebCodecs playback path [3]. During loading it enumerates packet metadata, removes packets with negative presentation timestamps, and sorts the remainder by presentation timestamp (PTS) and packet sequence number. Each entry stores PTS, duration, and whether the packet is a keyframe. Its position in this ordered list is the canonical zero-based frame index used by the player. Percent seeks are first quantized to that index; frame stepping then increments or decrements it directly, rather than accumulating floating-point time offsets.

Compressed video still cannot be decoded from an arbitrary inter-frame packet. For a cache miss, the player chooses a small window behind the requested frame, finds the preceding keyframe, restarts the decode iterator there, and advances until the target has been produced. Nearby forward seeks reuse the existing iterator, while backward or distant seeks restart when reuse would be invalid or more expensive. Decoded canvases are retained only in a bounded window

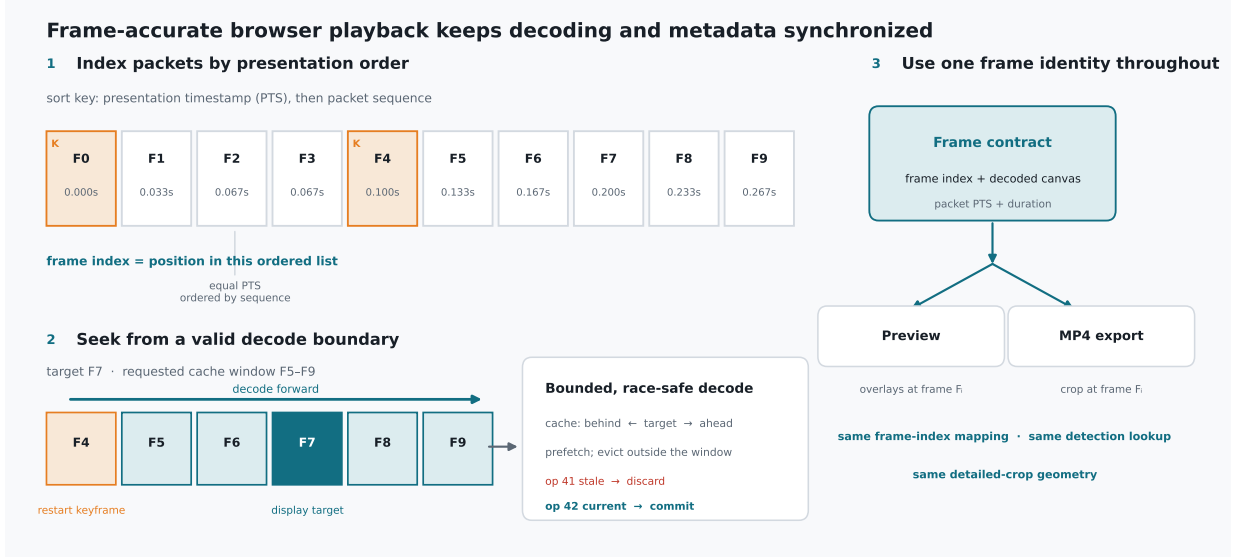


Figure 5: Frame-index contract for interactive playback and export. Packet metadata defines a deterministic presentation order; random access restarts at a preceding keyframe and decodes into a bounded, race-safe cache. Preview and export then select model metadata using the same frame identity and crop geometry.

around the current frame, with a short ahead region prefetched for playback. The cache improves interaction without making the complete decoded recording a memory requirement.

Asynchronous decoding introduces a less visible correctness problem: a slow result from an old seek can otherwise overwrite a newer user request. Every load or seek advances an operation identifier, and decode and prefetch work checks that identifier before publishing results. A changed operation makes prior work stale. Iterator and cache state are also reset when a forward-only iterator cannot serve the requested position, after disposal, and after returning from a hidden browser tab. These rules make backward stepping and rapid scrubbing deterministic at the application level, without claiming identical WebCodecs availability across all browsers.

The same ordered-packet convention is used when an export is decoded. Export samples are mapped to the corresponding packet index (with timestamp-based resynchronization if the iterators diverge), and that index selects the track detection used for the crop. Preview rendering likewise looks up detections and stabilization by the current frame index, and the detailed preview and exporter share the same crop routine. The result is one metadata contract from decoded image to model output to rendered crop.

This machinery is not a tracking contribution in itself; it is what makes the tracking result inspectable and exportable without silently moving its overlays to a neighboring frame.

6 Lessons and Limitations

This project is shaped less by the number of tracked objects than by the cost of a wrong association. The recordings usually contain only a handful of surfers, but they combine small targets, long overlaps and occlusions, changing backgrounds, and large camera pans. A single identity switch can make an otherwise complete focused video unusable. The appropriate operating point is therefore deliberately conservative: an uncertain continuation should remain fragmented or be rejected rather than silently merge two riders.

The resulting pipeline illustrates the value of matching representations to the task. The sail occupies more useful pixels than the rider and often carries a distinctive color pattern, so a compact sail-color descriptor can provide a more relevant association cue than generic person appearance features. Likewise, two semantic rig keypoints are sufficient to define a rider-relative

crop anchor and a physical scale signal; a full human-pose representation is unnecessary for this rendering objective. These components are modest individually, but encode the properties that remain observable in long-shot windsurfing footage.

Camera motion is a shared concern rather than an isolated stabilization step. The estimated background transform removes the dominant image motion before tracklet association and also supports stabilized presentation. Pose then solves a different rendering problem by stabilizing the composition of the selected rider. Treating tracking and rendering together avoids optimizing trajectories that cannot be delivered as visually stable, frame-consistent output.

Offline processing changes where ambiguity must be resolved. The local pass can form only high-confidence tracklets and defer uncertain crossings or gaps. The global association stage can then compare feasible continuations using observations before and after an occlusion, while retaining start, end, fragmentation, and discard as explicit alternatives to a weak link. Finally, frame-indexed decoding carries the resulting identity, pose, and stabilization metadata into browser preview and export without losing the frame correspondence on which those results depend.

The present evidence remains specific to this project. The repository does not yet contain a current, input-aligned benchmark against the implemented general-purpose trackers, so the report does not claim general or numerical superiority. The golden associations and retained diagnostics cover project footage and development cases; they do not establish performance across other venues, cameras, weather, or fleets. An aligned held-out comparison and complete-track audits would be required to measure how often the near-zero-switch objective is met.

Several ambiguities are also intrinsic to the chosen cues. Similar sail colors, illumination changes, spray, and contaminated crops can weaken the appearance descriptor. A long complete occlusion may admit multiple equally plausible global links even with future observations available. Pose localization degrades when the rig occupies too few pixels, forcing the virtual camera back towards less stable box geometry. Global motion estimation assumes that enough stable background texture remains visible after foreground masking; feature-poor water, motion blur, or newly revealed scenery during a pan can violate that assumption.

The main lesson is consequently not that custom machinery should replace general-purpose tracking. It is that a narrowly defined output and failure cost can justify a narrower system: conservative local decisions, domain-specific appearance and pose, camera-compensated motion, and non-causal global association. For this application, those choices make the system’s assumptions and failure modes explicit and align every stage with the identity-consistent focused video that the user ultimately inspects.

References

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